

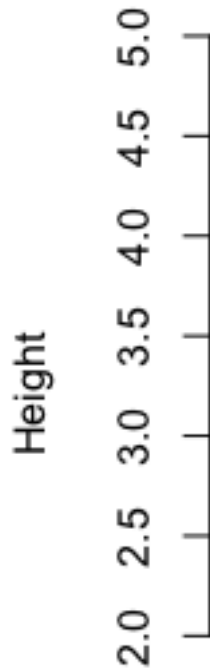
1. Hierarchical Clustering: **Total: 9 marks**

- (9 marks) (1) Suppose that we have five observations denoted by A, B, C, D, E , for which we have computed a dissimilarity matrix as follows.

	A	B	C	D	E
A		0.5	0.7	1.4	2.4
B	0.5		0.2	0.9	1.9
C	0.7	0.2		0.7	1.7
D	1.4	0.9	0.7		1
E	2.4	1.9	1.7	1	

Based on this dissimilarity matrix, you are required to perform **agglomerative (hierarchical) clustering** on the above five observations using **single linkage**. To finish this question, you need to

- write down the steps that you use in details to cluster the five observations, according to the hierarchical clustering algorithm;
- sketch the dendrogram, and make sure that you indicate on the plot the height at which each fusion occurs, as well as the observations corresponding to each leaf in the dendrogram.



Solution: We will use Algorithm 12.4 *Hierarchical Clustering* to explain the detailed steps leading to the dendrogram.

Step 0 We already have

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>
<i>A</i>		0.5	0.7	1.4	2.4
<i>B</i>	0.5		0.2	0.9	1.9
<i>C</i>	0.7	0.2		0.7	1.7
<i>D</i>	1.4	0.9	0.7		1
<i>E</i>	2.4	1.9	1.7	1	

Step 1 For the clusters $i = \{1, \dots, 5\}$,

Step 1 (a) : We examine all pairwise dissimilarities and see that 0.2 is the minimum dissimilarity, so we fuse observations *B* and *C* to form a cluster (B, C) at height 0.2. (0.5 mark)

Step 1 (b) : Now, we use single linkage to compute the dissimilarity

- between (B, C) and *A*: (0.5 mark)

$$\min(\text{dis}(A, B), \text{dis}(A, C)) = \min(0.5, 0.7) = 0.5$$

- between (B, C) and *D* (0.5 mark)

$$\min(\text{dis}(B, D), \text{dis}(C, D)) = \min(0.9, 0.7) = 0.7$$

- between (B, C) and *E* (0.5 mark)

$$\min(\text{dis}(B, E), \text{dis}(C, E)) = \min(1.9, 1.7) = 1.7$$

So, we have the following new dissimilarity matrix: (0.5 mark)

	<i>A</i>	(B, C)	<i>D</i>	<i>E</i>
<i>A</i>		0.5	1.4	2.4
(B, C)	0.5		0.7	1.7
<i>D</i>	1.4	0.7		1
<i>E</i>	2.4	1.7	1	

Step 2 For the 4 remaining clusters,

Step 2(a) : We see that the minimum dissimilarity is 0.5 between (B, C) and *A*, so we fuse cluster (B, C) and observation *A* to form a cluster $(A, (B, C))$ at height 0.5. (0.5 mark)

Step 2(b) : Next, we compute the dissimilarity between $(A, (B, C))$ and *D*, *E*, respectively using single linkage in a similar way.

- between $(A, (B, C))$ and *D* (0.5 mark)

$$\min(\text{dis}((B, C), D), \text{dis}(A, D)) = \min(0.7, 1.4) = 0.7$$

- between $(A, (B, C))$ and *E* (0.5 mark)

$$\min(\text{dis}((B, C), E), \text{dis}(A, E)) = \min(1.7, 2.4) = 1.7$$

We now have the new dissimilarity matrix (0.5 mark)

	$(A, (B, C))$	D	E
$(A, (B, C))$		0.7	1.7
D	0.7		1
E	1.7	1	

Step 3 For the 3 remaining clusters,

Step 3(a) : The minimum dissimilarity is 0.7, so we merge $(A, (B, C))$ and D to form a new cluster $((A, (B, C)), D)$ at height 0.7 (**0.5 mark**)

Step 3(b) : Next, we compute the distance between $((A, (B, C)), D)$ and E as (**0.5 mark**)

$$\begin{aligned} & \min(\text{dis}((A, (B, C)), E), \text{dis}(D, E)) \\ &= \min(1.7, 1) = 1 \end{aligned}$$

Step 4 For the 2 remaining clusters, finally, we merge $((A, (B, C)), D), E)$ at 1.

Single Linkage

